

Ivan Kashuba

ivan.med.tr@gmail.com

Research Engineer | Chemist – Biopolymer & Coating Technologies

Professional Summary

Research engineer and technologist with 10+ years of hands-on experience. Experience includes growing and processing single crystals, extruding optical fibers, synthesizing monomers for biopolymers, and selecting one powder coating components. Committed to continuous learning and scientific advancement. Seeking research opportunities in academic and industrial environments.

Key Skills and Technical Tools

- Analytics: NMR, DSC, TGA, DTA, MS
 - Software: DSC, MestReNova, Chem3D, ChemDraw, neural networks
 - Equipment: microwave reactor, extrusion tools, DSC (Mettler), NMR (Bruker), spectrophotometers, FTIR, creating vacuum, grinding, heating equipment, laboratory drying equipment, powder moisture meter, electron-beam sputtering, chemical laboratory equipment, CO₂-laser, potentiostat, conductometer, pH meter, thickness gauges, UV lamp, thermocouples, ultrasonic dispersing bath, centrifuge
 - Soft skills: working with new information, data interpretation, critical thinking
 - Remote Collaboration: Zoom, Google Meet, Microsoft Teams, Slack, Trello, Asana, Notion (if used)
 - Documentation & Presentation: Google Docs, Google Slides, MS Office, Adobe, training manuals
- Basic Skills (currently improving): python scripting by neural networks, molecular docking tools
- Scientific Databases & Research Tools: Google Scholar, Google Colab, Reaxys

Education

Ural Federal University named after the First President of Russia B.N. Yeltsin, Ekaterinburg (UrFU) 2015-2019

Institute of Chemical Engineering, department of Physical Chemistry and Chemistry of Colloids. Postgraduate course in the direction Chemical Sciences. Qualification: «Researcher. Teacher-researcher». Thesis: «Research and creation of photonic crystal light guides for a wide IR spectral range». GPA: 4.4/5.0. Refined the AgBr–AgI phase diagram using DTA for creation of single-mode light guides with a wider IR transmission range for laser surgery. Together with the scientific director and the team, synthesized a charge using thermo-zone crystallization, which allows to synthesis of purify low-soluble silver halides with a high degree of purity and correctness of the crystal lattice due to low supersaturation of the solution. We grew single crystals the Bridgman-Stockbarger method. Extruded light guides with a wider spectral range and degradable by UV. Next step - increase photostability. Turned about a dozen different single crystals on a small lathe in a glove box to the required dimensions of the extrusion tooling. Extruded them with heating into light guides. Participated in these articles: <https://orcid.org/0009-0002-8976-8886> / [https://sciencedata.urfu.ru/portal/en/persons/--\(29b208fc-6d95-4f7b-9cdd-c0fb239d9e02\)/publications.html](https://sciencedata.urfu.ru/portal/en/persons/--(29b208fc-6d95-4f7b-9cdd-c0fb239d9e02)/publications.html)

Ural Federal University (UrFU)

2010-2015

Institute of Chemical Engineering. Specialization: Chemical technology of single crystals, materials and electronic products. Thesis: «Study of extrusion modes of infrared light guides based on silver halides monocrystals». Specialist degree. GPA: 4.4/5.0. Researched extrusion modes for optical fibers from two-component silver halide systems and its optic parameters. Modeled and tested hot extrusion tooling in ANSYS for strength. Studied the hot extrusion modes of IR light guides. Designed, calculated and built a tubular furnace for hot extrusion of single crystals in the Shimadzu Tensile Testing Machine AGS-X press. Processed the single crystals on a small lathe in a glove box. Used Origin and learn LaTeX.

UrFU Business School

2012-2015

Specialty: Economics and business management. Together with my colleagues calculated the profitability of a scientific and production project.

Professional Interests. Study and apply scientific knowledge, articles, technologies for useful working solutions to modern problems. I am interested in expanding fundamental scientific knowledge and sharing it.

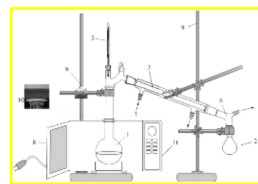
Professional development

UrFU, 2023-2024, specialty: Resource-saving methods and technologies of functional materials and biologically active substances. Learned the basics of operating an NMR spectrometer and interpreting the spectra of H¹, C¹³ and other isotopes, as well as the basics of reading two-dimensional spectra. Learned making equations from similarity criteria by excel to scale biotechnological processes. Studied TLC, HPLC, LC-MS, GC-MS, UPLC, IR, UV-Vis. Identified binding sites using proteins.plus, PyMOL, AutoDockTools, www.ncbi.nlm.nih.gov, cheminfo.org, pubmed.ncbi.nlm.nih.gov, OpenBabel, [SwissTargetPrediction](https://www.swisstargetprediction.ch), rscb.org.

Explored methods to increase the yield of monomer synthesis for obtaining biodegradable polymers by microwave heating and using a zeolite catalyst. Selection of a heated medium. Designed a microwave reactor for monomer synthesis based on literature review. It is universal for synthesis from substances sensitive to the effects of radiation in the microwave range at which microwave oven magnetrons operate, and for vacuum distillation of products.

Calculated the microwave resonance energy transfer of liquids according to the hypothetical calculation by wavelengths and molecular sizes. Calculated interatomic distances using Python Molecule Viewer. This was implemented during project work at UrFU in collaboration with the monomer laboratory Medin-N.

Studied creating neural networks in python with libraries plugged in.



Professional experience

Research engineer in Institute for Applied Chemistry and Certification FRISCHBERG Ltd.,

July 2024 - January 2025, 5.5 months

- Tested the replacement of the main components of mass-produced powder coating on laboratory equipment
- ✓ Installation of some equipment in the laboratory for the development and production of powder paints has been planned and completed
 - ✓ Studied the technical documentation for the equipment in English and successfully applied it in development. Together with the team, a DSC with a nitrogen cylinder and a cooler was installed. Measured the reaction enthalpy and thermal transitions of powder coatings
 - ✓ Inspected and cleaned the extruder heat exchanger to optimize zone temperature control. Alkalinisation with aliphatic amine and adding an anti-corrosion additive to the heat transfer medium
- ✓ Improving the quality of some of the powder by sealing it during storage and cleaning the micronization system
- ✓ By measuring the gelation time of the multi-component coating also selected replacement components
- ✓ Improving the homogeneity of a powder mixture by grinding new components and sifting them through a sieve
- ✓ Processed and visualized data in Excel for calculating the dependence of gelation time on the content of components, the diameter of cathodic peeling on the humidity of powders, calculating the weights and for detecting correlations
- ✓ Selected resin solvents for recording NMR spectra, thanks to which, together with the supervisor selected composition
- ✓ Performed qualitative reactions: for iron - with alkali, for pyrocatechol, i.e. hydroxyl groups in the ortho position - with an alkaline reaction and with iron chloride, soda and ammonia, for iodine - with a reaction with iodine and starch
- ✓ Tribostatically sprayed powders into a ventilation chamber onto hot test specimens. As a result of the work, good adhesion of the coating and a good diameter of cathodic peeling were obtained. Produced about 30 kg of powder coatings for experiments.

Chemist-technologist in Medin-N, August 2021 - April 2024, 33 months

- Maintaining and improving the quality of the monomer 1,4-dioxane-2,5-dione
 - ✓ In the summer the monomer became cleaner from water and more often passed quality control
 - ✓ Found the inefficiency of raw material dehydration and a modern drying cabinet was installed
 - ✓ Oxidation of monochloroacetic acid (raw material) during storage caused by depressurization of pH measurement bags was observed and eliminated, which improved the quality and quantity of monomer produced
 - ✓ Found the range of temperature modes for the operation of a thermal vacuum reactor for the synthesis and distillation of 1,4-dioxane-2,5-dione
 - ✓ Improved the filtration of glycolide solution from oligomers by insulating the filter element, which corrected the need to overheat the solution and dissolve some of the oligomers before filtration
- All this increased the quality and amount of manufactured products. Biodegradable polymers are the main components of surgical absorbable biocompatible sutures and pins.
- ✓ Synthesized about 600 kg of monomers
 - Studied and solved the problem of increasing and stabilizing the quality of a protective solution of dye with a polymer that crosslinks when heated. Ground and heated the initial components under stirring for complete dissolution. Instead of 2 weeks, the solution can now be prepared in a few hours
 - Together with the monomer laboratory, for project work at UrFU, made a setup for microwave synthesis and distillation. Made ¹H NMR monomer, which allowed me to see its rapid oligomerization due to atmospheric moisture

Process engineer in OOO ODV Prescription Optics Plant, April 2021 - July 2021, 4 months

- Studied and supported the technology of washing and increasing the adhesion of the surface of polymer lenses. Application of optical interference coatings using old and new equipment for cleaning lenses and vacuum sputtering units with electron-beam emission of the original components
- ✓ Removed the hardened flaking polymer from the finishing zone
- ✓ Transferred the recipe for interference coating to new electron-beam sputtering equipment at low pressure, using a formula given by the director. Measured the moisture content of the polymer used and the transmission spectra of coatings on spectrophotometer
- ✓ Maintained Excel logs and resolved coating defects promptly resolving lens coating defects using a microscope, humidity sensor, temperature, and equipment operating parameters

Research engineer in Science Lab of Fiber Technology and Photonics, Center for IR Fiber Technology early, January 2013 - July 2021, 8 years 8 months

- Together with scientific director and team, participated in the development and production of a flexible neonatal lamp based on PMMA light guides for the treatment of hyperbilirubinemia in premature infants
 - ✓ Machined connectors and attachment of fibers
 - ✓ Suggested automatically laser engraving light guides to create side glow and found where to buy the equipment
 - ✓ With the team, assembled an LED lamp for fast curing of UV glue for fixing connectors
- Co-developed and implemented solutions over 3 years with successful outcomes.



Participated in volunteer projects supporting children in orphanages and am particularly interested in medical applications of science research

Languages

Russian – native. English B2 Fluency. Chinese A1 – Elementary